

RADWAY AGCM, AB, Canada

WMO: 715450

Lat: 53.9950N

Lon: 112.8791W

Elev: 625

StdP: 94.04

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.2	-29.7	-36.8	0.1	-33.1	-33.2	0.2	-29.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	28.1	17.3	26.4	16.6	24.8	16.0	19.4	25.3	18.1	24.0	17.1	22.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	13.2	22.7	15.9	12.2	20.9	14.8	11.3	19.7	58.0	25.5	53.7	24.1	50.3	22.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-38.1	31.4	3.8	0.9	-40.8	32.0	-43.1	32.5	-45.2	33.0	-48.0	33.7	(4)
(5)			WB	-38.2	21.6	3.8	1.9	-40.9	22.9	-43.1	24.0	-45.3	25.1	-48.0	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-11.5	-11.9	-5.5	3.2	10.8	14.6	16.7	15.6	10.9	3.7	-5.2	-12.5	(6)
(7)		DBStd	12.40	8.71	8.30	8.01	5.26	4.56	2.97	2.47	3.16	4.43	4.68	6.87	7.93	(7)
(8)		HDD10.0	3416	667	614	481	209	46	3	0	1	42	199	456	698	(8)
(9)		HDD18.3	5812	925	848	739	453	234	114	63	95	226	452	706	957	(9)
(10)		CDD10.0	674	0	0	0	6	72	143	208	174	67	6	0	0	(10)
(11)		CDD18.3	28	0	0	0	0	2	4	12	9	1	0	0	0	(11)
(12)		CDH23.3	706	0	0	0	3	85	93	216	208	99	2	0	0	(12)
(13)		CDH26.7	129	0	0	0	0	16	12	36	42	24	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	431	22	12	18	34	48	75	85	53	36	21	22	18	(15)
(16)		PrecMax	606	89	25	47	64	123	146	161	99	86	55	89	41	(16)
(17)		PrecMin	248	3	1	6	0	8	25	22	11	6	1	2	4	(17)
(18)		PrecStd	70	19	7	9	17	24	33	30	22	21	13	18	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.7	12.6	22.9	28.9	28.1	29.8	30.2	29.9	22.1	11.7	5.7	(19)
(20)			MCWB	2.9	3.0	6.6	10.6	14.1	16.9	19.6	17.6	16.5	11.9	6.4	2.0	(20)
(21)		2%	DB	4.7	3.9	8.6	18.5	25.7	26.0	27.5	27.6	26.2	17.0	7.6	3.2	(21)
(22)			MCWB	1.9	1.2	3.9	8.3	12.4	16.1	18.8	17.9	15.0	9.4	4.1	0.4	(22)
(23)		5%	DB	3.1	2.0	6.2	15.3	23.2	23.9	25.8	25.5	23.3	14.0	5.3	1.1	(23)
(24)			MCWB	0.9	0.0	2.6	7.0	11.6	14.6	18.0	17.1	13.8	7.4	2.3	-1.0	(24)
(25)		10%	DB	1.2	0.2	4.1	12.2	20.8	22.1	24.0	23.8	20.0	11.5	3.2	-1.4	(25)
(26)			MCWB	-0.6	-1.4	1.3	5.4	10.5	14.0	16.8	16.4	12.4	6.4	1.3	-2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.3	6.9	11.1	14.7	18.9	21.7	20.6	17.3	12.5	6.9	2.5	(27)
(28)			MCDWB	6.1	6.5	11.6	22.5	24.8	25.0	26.7	26.1	26.2	21.1	10.6	5.4	(28)
(29)		2%	WB	2.1	1.5	4.5	8.8	13.5	17.2	19.9	19.2	15.7	10.1	4.5	0.6	(29)
(30)			MCDWB	4.3	3.4	8.3	17.7	22.5	22.9	25.8	25.3	24.7	15.3	7.2	3.0	(30)
(31)		5%	WB	1.0	0.2	2.8	7.2	12.5	16.0	18.6	17.8	14.3	8.3	2.8	-1.1	(31)
(32)			MCDWB	3.0	1.8	6.1	14.3	21.4	22.4	24.2	23.6	22.2	12.9	4.9	0.9	(32)
(33)		10%	WB	-0.6	-1.5	1.4	5.9	11.4	15.0	17.5	16.8	13.0	6.9	1.2	-2.8	(33)
(34)			MCDWB	1.2	0.2	3.8	11.3	19.2	20.6	22.8	22.6	19.1	10.9	3.0	-1.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	11.2	10.6	11.6	14.1	12.1	12.6	13.6	14.0	11.2	8.9	9.4	(35)
(36)			MCDBR	10.7	12.3	12.1	17.4	18.5	15.5	15.9	17.1	20.6	15.8	11.0	11.0	(36)
(37)			MCWBR	8.6	10.0	8.1	8.8	7.9	7.2	8.4	8.3	9.9	9.0	7.6	9.0	(37)
(38)		5% WB	MCDBR	10.4	11.5	11.6	15.9	16.7	13.2	14.5	15.0	19.0	15.1	10.2	11.2	(38)
(39)			MCWBR	8.4	9.5	7.9	8.4	7.3	6.7	8.2	7.9	9.4	9.0	7.5	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.267	0.310	0.329	0.342	0.356	0.355	0.364	0.308	0.281	0.252	0.241	(40)	
(41)		taud	2.291	2.327	2.292	2.374	2.391	2.367	2.414	2.342	2.501	2.548	2.448	2.280	(41)	
(42)		Ebn at Noon	721	843	870	901	904	891	885	853	873	830	742	663	(42)	
(43)		Edh at Noon	56	76	100	106	110	114	107	108	81	62	48	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.91	1.94	3.47	4.66	5.70	5.98	6.08	4.99	3.44	1.96	0.92	0.66	(44)	
(45)		RadStd	0.10	0.14	0.26	0.41	0.45	0.40	0.31	0.35	0.35	0.25	0.10	0.06	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page